

# Reliability-Aware Hybrid Retrieval for Natural-Language-to-Shell-Command Assistance: A Non-LLM Study

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## Abstract

Natural-language interfaces to the shell are usually framed as generation: translate an English request into a novel command (NL2Bash, NL2SH). We study a deliberately constrained alternative — closed-vocabulary *retrieval* from a curated, pre-vetted command library, with no large language model anywhere in the system — and focus on *reliability* rather than accuracy alone. Starting from a real, previously shipped BM25 baseline (67.3% overall accuracy on a 150-query benchmark, but 86.06% mean confidence on its wrong answers), we test whether a lightweight hybrid lexical+local-semantic retriever with post-hoc isotonic calibration and margin-based rejection improves accuracy and reliability, under leakage-free nested 5-fold cross-validation. We report two benchmark versions: v0.1 (150 queries) and v0.2, which extends v0.1 with 35 out-of-domain (OOD) and 24 ambiguous queries to increase coverage of those cases; the versions therefore share their 150 queries and are not independent samples. Calibration is the most robust result: relative ECE falls by 56.7–84.1%, it is the only one of four compared effects significant after Holm–Bonferroni correction on both versions (adjusted  $p = 0.0004$  and  $0.0003$ ), and it persists on intents held out from tuning (69–77% vs. 76–80%) and when the ambiguous queries are removed. OOD rejection, underpowered at 15 queries (raw  $p = 0.25$ ), is significant on v0.2’s 50 OOD queries (34.0%→68.0%; adjusted  $p = 0.00006$ ). The hybrid’s accuracy gain over BM25 is consistent in size (+5.2 and +3.8 points) but statistically fragile: adjusted  $p = 0.047$  on v0.1 and raw  $p = 0.070$  on v0.2, where one bare-keyword query is the only one BM25 wins; the hybrid’s advantage over dense retrieval on v0.2 (adjusted  $p = 0.025$ ) disappears when bare-keyword ambiguous queries are excluded (raw  $p = 0.057$ ). Ambiguity detection remains weak (F1 0.31–0.50). A rule-based safety classifier reaches 95%/95% precision/recall with no dangerous-direction

misses. A blind re-annotation of 14 of the 59 AI-authored v0.2 queries by one partially independent reviewer gave Cohen’s  $\kappa = 0.63$ , below our 0.7 target: OOD labels agreed fully (8/8), and all disagreements were on ambiguous labels (3/6). The significance analyses were specified after the raw results were seen and are exploratory. We report all null, weak, and non-robust results.

## 1 Introduction

Natural-language shell assistance can lower the barrier to command-line tools, but incorrect commands — especially destructive ones — carry real cost. The dominant research trajectory since NL2Bash (Lin et al., 2018) has been generative: seq2seq and Transformer models (the NLC2CMD competition, Agarwal et al., 2021), then LLM-based systems (NL2SH; Westenfelder et al., 2025), most recently with execution-grounded, RL-trained approaches (BashCoder-R1; Yu et al., 2026). This trajectory inherits a persistent risk: open-vocabulary generation can hallucinate syntactically valid but semantically wrong or unsafe commands.

We study a different, narrower approach implemented in a real, previously-shipped, in-use npm package: retrieval from a fixed, pre-vetted corpus of 431 records spanning 279 unique command intents on the evaluated platform, using classical lexical (BM25) scoring, with no LLM anywhere in the system. This constrains coverage but structurally eliminates hallucination of arbitrary commands. We audited and reproduced the system’s baseline behavior (§3) and found a specific, serious problem: the system is frequently *confidently* wrong, not merely wrong. This paper asks whether that problem, and the system’s accuracy more broadly, can be improved without abandoning the no-LLM, fully-offline design constraint.

Given a natural-language query and a fixed corpus of (intent, command, category, description, os) tuples, the task is to return

the single best-matching command, or explicitly decline to answer if no corpus entry is a confident match — retrieval, not generation, over a closed and pre-vetted output space, unlike NL2Bash-style tasks where the model may emit any syntactically valid command string.

**Research question.** Can a lightweight hybrid lexical+local-semantic retrieval architecture, combined with explicit uncertainty estimation and post-hoc calibration, measurably improve accuracy and reliability over a pure-BM25 baseline, without the cost or hallucination risk of a generative system?

**Contributions.** (1) A reproduced audit of a real shipped BM25 baseline’s accuracy and — more consequentially — its confidence miscalibration. (2) A five-component reliability-aware retrieval architecture evaluated under leakage-free nested cross-validation. (3) Results on two benchmark versions (v0.2 extends v0.1, so they are not independent), with Holm–Bonferroni-corrected significance and a sensitivity analysis showing which conclusions depend on bare-keyword ambiguous queries. (4) An intent-held-out (GroupKFold) generalization check. (5) Disclosure of the null, weak, and fragile results we found, including a below-target blind re-annotation of a sample of our own benchmark’s AI-authored queries.

## 2 Related Work

NL2Bash (Lin et al., 2018) established the seq2seq/CopyNet/Tellina baseline and the BLEU/template/exact-match evaluation this field has used since. The NLC2CMD competition (Agarwal et al., 2021) introduced a utility+flag-overlap metric more tolerant of near-miss answers than exact match. NL2SH (Westenfelder et al., 2025) is the modern generation-based successor, with even SOTA LLM accuracy reported as “remains low.” Recent work reinforces our framing that exact-match accuracy alone is an incomplete signal: QuoteBench (Li et al., 2026) shows matched execution scores can mask command-path failures, and BashCoder-R1 (Yu et al., 2026), despite RL training against execution feedback, tops out at 73–90% full functional success even as a strong system. A 2026 scaling study (Wang et al., 2026) finds BM25 overtakes dense and agentic retrieval past roughly 10M corpus tokens, supporting lexical retrieval’s competitiveness at scale, though at a corpus scale far larger than ours.

Notaro et al. (2024) train a transformer-based command-risk classifier; we deliberately compare against a simpler, fully deterministic rule-based alternative (§7). Our study complements this generation-focused work with a reliability-oriented evaluation of closed-vocabulary retrieval: calibration, OOD rejection, and intent-held-out evaluation of a lexical/hybrid retriever. We do not claim priority; our literature search was limited (see Limitations).

## 3 TermAssist: Baseline System and Benchmark

**Baseline architecture** (frozen for this study): tokenize (lowercase, strip punctuation, 10-word stopword list), BM25 scoring ( $k_1 = 1.2$ ,  $b = 0.75$ , standard IDF smoothing) over intent+category+description, plus a flat +15.0 bonus when the query is a near-substring of the command’s intent. Confidence =  $\min(\text{round}(\text{score}/8 \times 100), 100)$ , rejected below score < 2.0. 431 corpus records, 27 categories, platform-filtered (win32/darwin/linux/all). No embeddings, no LLM, mean latency 3.29ms. We reproduced this baseline’s outputs exactly (0/150 mismatches against the archived results). **Corpus size clarification:** 134 of the 431 records are cross-platform duplicates of the same intent, so the retrieval space actually exercised on the evaluated (win32) platform is **279 unique intents, not 431** — the number we use throughout when comparing corpus scale to other work. A planned expansion to 500–800 human-validated intents requires human authoring and validation and is not complete, so all results here are on the original 279-intent corpus.

**Benchmark, v0.1.** 150 hand-adjudicated queries across 9 types (canonical, paraphrase, low-overlap-paraphrase, ambiguous, OOD, polysemy, single-keyword, safety-sensitive, complex-multi-intent), with ground-truth classification (119 CORRECT, 14 AMBIGUOUS, 15 OOD, 2 NEEDS\_CORRECTION) and a risk-level field (120 LOW, 10 MEDIUM, 15 HIGH, 5 CRITICAL) reused in the safety evaluation (§7).

**Benchmark, v0.2.** A second version (209 queries) *extends* v0.1: it contains all 150 v0.1 queries plus 35 new OOD and 24 new ambiguous queries, added because v0.1’s OOD (15) and ambiguous (14) subsets were too small for adequately powered tests. It is therefore an expanded

benchmark, not an independent sample: the 121 answerable queries are identical in both versions, and every difference between the versions comes from the added OOD and ambiguous queries. The new queries were authored and adjudicated by an AI agent under human direction, following v0.1’s design methodology (each query checked against actual system retrieval output; 6 of 30 drafted ambiguous candidates were rejected for failing the ambiguity or novelty criteria). Seventeen of the 24 new ambiguous queries are single-token tool or category names (“grep”, “pip”, “system”); we call these *bare-keyword* queries, and nine of v0.1’s 14 ambiguous queries are of the same kind (“git”, “docker”).

**Baseline results (v0.1).** Overall accuracy 67.3% (101/150); supported-task (non-OD) accuracy 71.9%; OOD rejection 26.7% (73.3% false-acceptance); ambiguity success 28.6%. By type: canonical/safety-sensitive 100%, paraphrase 88.0%, complex-multi-intent 86.7%, polysemy 55.6%, ambiguous 58.3%, low-overlap-paraphrase 33.3%, OOD 26.7%, single-keyword 0.0%. Mean confidence on the 49 wrong (non-rejected) predictions: **86.06%**, with 44.9% of failures at exactly 100% confidence — the central problem motivating this paper.

## 4 Reliability-Aware Retrieval Architecture

Five components, each targeting a specific measured failure: (1) local dense retrieval (Xenova/all-MiniLM-L6-v2, 384-dim, of-line after one-time download) for the lexical gap; (2) hybrid score fusion with  $\alpha$  tuned via nested 5-fold CV; (3) margin/entropy uncertainty signals over top-10 ranked candidates; (4) OOD and ambiguity detectors (dev-tuned thresholds); (5) isotonic confidence calibration. A sixth, independent component — a rule-based safety classifier — targets command risk, orthogonal to retrieval correctness. No LLM is used anywhere in the system.

## 5 Experimental Methodology

All tuned parameters (fusion  $\alpha$ , OOD/ambiguity thresholds) are selected via nested 5-fold cross-validation (seed 42, stratified by classification) — **Split A:** for each test fold, the parameter is chosen using only the other four folds, then applied once to the held-out fold, never tuned on data it is later scored against. Statistical testing uses the exact

(binomial) McNemar’s test, given small discordant-pair counts.

**Multiple-comparison correction.** We fix a family of four comparisons per benchmark version — accuracy vs. BM25, accuracy vs. dense-alone, OOD rejection, and calibration ECE reduction — and apply Holm–Bonferroni correction within each family; a result is called significant only at Holm-adjusted  $p < 0.05$ . *This family was written down after the raw v0.1 and v0.2 test results had been computed* (only the Holm-adjusted values and bootstrap intervals came later), and calibration was named the primary hypothesis knowing it was the robust result. All significance statements here are therefore exploratory with respect to these data; the specification would be confirmatory only on a benchmark built after it. The two families are also not independent, since v0.2 contains v0.1.

**Bootstrap confidence intervals.** Percentile bootstrap (10,000 resamples, seed 42) provides 95% CIs for the accuracy deltas (paired resampling over non-OD query ids) and for the calibration ECE reduction (resampling over pooled held-out predictions), complementing the discrete exact McNemar’s test with a continuous-estimate view of the same comparisons.

**Split B — intent-held-out generalization.** To test whether Split A’s findings depend on the same command intents appearing in both tuning and test folds, we additionally construct a GroupKFold split keyed by a derived intent-group id (the gold command for answerable queries; the sorted acceptable-command set for ambiguous queries; a singleton per query for OOD queries, which target no intent). We run this on both benchmark versions (171 groups over v0.2’s 209 queries; 125 groups over v0.1’s 150 queries), both verified programmatically to place zero groups across more than one fold, so  $\alpha$ , the detector thresholds, and the isotonic calibrator are always selected on queries targeting *different* command intents than those they are scored on.

## 6 Results

**Retrieval accuracy.** On v0.1, BM25 reaches 71.9% on the 135 non-OD queries, dense-only 72.6–72.7%, and hybrid 77.1% (bootstrap 95% CI on the BM25→hybrid delta [2.2, 8.9]pp; exact McNemar  $p = 0.0156$ , Holm-adjusted  $p = 0.047$ :

significant, but by 0.003). On v0.2 (159 non-  
 OOD queries), BM25 reaches 75.4%, dense-only  
 72.3–72.4%, and hybrid 79.3%: a gain of similar  
 size (+3.8pp, CI [0.6, 7.5]pp) that does not clear  
 $p < 0.05$  even before correction (raw  $p = 0.070$ ).  
 The exact test rests on very few discordant queries:  
 7 on v0.1 (all favoring hybrid) and 8 on v0.2 (7  
 favoring hybrid). The single query on which BM25  
 succeeds and hybrid fails is the bare keyword “sys-  
 tem” (TA-B194); without it the v0.2 comparison  
 is 7–0, exactly as on v0.1. The accuracy gain is  
 therefore consistent in size across versions, but its  
 significance depends on one query, and we read  
 it as suggestive rather than established. Hybrid  
 vs. dense-alone is not significant on v0.1 (Holm-  
 adjusted  $p = 0.292$ ) but is on v0.2 (Holm-adjusted  
 $p = 0.025$ ); three of the 14 queries on which hy-  
 brid beats dense are new bare-keyword queries (jq,  
 grep, shell), and without them the comparison  
 is 3–11 (raw  $p = 0.057$ ). We suspect exact-token  
 matching helps on such queries, but we did not test  
 this.

By query type (v0.1), hybrid retains BM25’s  
 100% on canonical and safety-sensitive queries  
 while improving most on low-overlap-paraphrase;  
 single-keyword and OOD queries remain the hard-  
 est categories for both systems, and v0.2 has a  
 similar shape (Figure 3, Appendix A).

**OOD and ambiguity detection.** On v0.1: AU-  
 ROC 0.867 (OOD, feature = absolute top-1 score)  
 and 0.784 (ambiguity, feature = margin) — an  
 evidence-driven feature choice (§4), not selected  
 post-hoc. On v0.2 (OOD subset expanded 15→50  
 queries): OOD AUROC improves to 0.901,  
 and the baseline-vs-tuned rejection-rate improve-  
 ment (34.0%→68.0%) **survives Holm correction**  
 (Holm-adjusted  $p = 0.00006$ ) — resolving the  
 power limitation of the v0.1-scale result, which  
 was not significant even before correction (raw  
 $p = 0.25$ ). Ambiguity detection F1 improves on  
 v0.2 (0.310→0.496) though AUROC slightly de-  
 creases (0.784→0.723), reported as measured.

**Calibration.** ECE is reduced 56.7–84.1% across  
 four confidence variants tested, largest for the hy-  
 brid system’s own fused-score signal. A paired  
 bootstrap test of the reduction gives one-sided  
 $p \approx 0.0001$  on both benchmarks, and this is the  
**only comparison in the primary family that sur-  
 vives Holm correction on both benchmark ver-  
 sions** (Table 1). On v0.1, isotonic calibration re-  
 duces the hybrid system’s own reliability-diagram

| BM   | Comparison               | Raw $p$  | Holm    | Sig. |
|------|--------------------------|----------|---------|------|
| v0.1 | Acc., hybrid vs. BM25    | 0.0156   | 0.0469  | Yes  |
| v0.1 | Acc., hybrid vs. dense   | 0.1460   | 0.2920  | No   |
| v0.1 | OOD rej., tuned vs. base | 0.25     | 0.2920  | No   |
| v0.1 | Calib. ECE reduction     | 0.0001   | 0.0004  | Yes  |
| v0.2 | Acc., hybrid vs. BM25    | 0.0703   | 0.0703  | No   |
| v0.2 | Acc., hybrid vs. dense   | 0.0127   | 0.0255  | Yes  |
| v0.2 | OOD rej., tuned vs. base | 0.000015 | 0.00006 | Yes  |
| v0.2 | Calib. ECE reduction     | 0.0001   | 0.0003  | Yes  |

Table 1: Holm-corrected significance ( $\alpha = 0.05$ ; ex-  
 ploratory, see §5). Bootstrap 95% CIs: accuracy vs.  
 BM25 [2.2, 8.9]pp (v0.1) and [0.6, 7.5]pp (v0.2); ac-  
 curacy vs. dense [−0.7, 9.6]pp and [1.9, 11.9]pp; ECE  
 reduction [0.128, 0.283] and [0.172, 0.295].

ECE from 0.2741 to 0.0537 (Figure 2, left); on  
 v0.2 the same signal’s ECE reduces from 0.3237 to  
 0.0738 (Figure 2, right) — both large reductions,  
 though the v0.2 starting ECE is itself higher, consis-  
 tent with the harder query mix discussed through-  
 out this section. Selective answering at 50% cover-  
 age achieves 10.7% error vs. 30.7% unconditional  
 on v0.1; risk-coverage curves for both versions are  
 in Figure 4 (Appendix A).

**Sensitivity to bare-keyword queries.** A blind  
 reviewer disagreed with our labels on several short  
 ambiguous queries (§8), so we re-evaluated on  
 subsets that drop them (Table 2). The rule was  
 fixed before looking at outcomes (an ambiguous-  
 labeled query with a single token, applied to both  
 versions), but the analysis itself is post hoc and  
 exploratory; the fusion weight and detector thresh-  
 olds were *not* re-tuned on the reduced sets, while  
 calibration was re-fit. Three results stand out. (i)  
 The BM25→hybrid gain is stable in size, +4.9 to  
 +5.8 points on every reduced subset, each with  
 7–0 discordant queries in hybrid’s favor. (ii) The  
 hybrid-over-dense advantage on v0.2 is not stable:  
 it falls to raw  $p = 0.057$  without the new bare-  
 keyword queries and to  $p = 0.227$  on answerable  
 queries only. (iii) The calibration gain is unchanged  
 (77–84% relative ECE reduction across all rows,  
 one-sided bootstrap  $p \leq 0.0001$  in each). OOD  
 rejection is not recomputed: relabeling ambiguous  
 queries cannot change it, because ambiguous and  
 answerable queries are both non-*OOD* negatives  
 and the OOD queries are identical in every sub-  
 set. Finally, on answerable queries only, v0.1 and  
 v0.2 are the same 121 queries, so v0.2 adds no  
 answerable evidence.

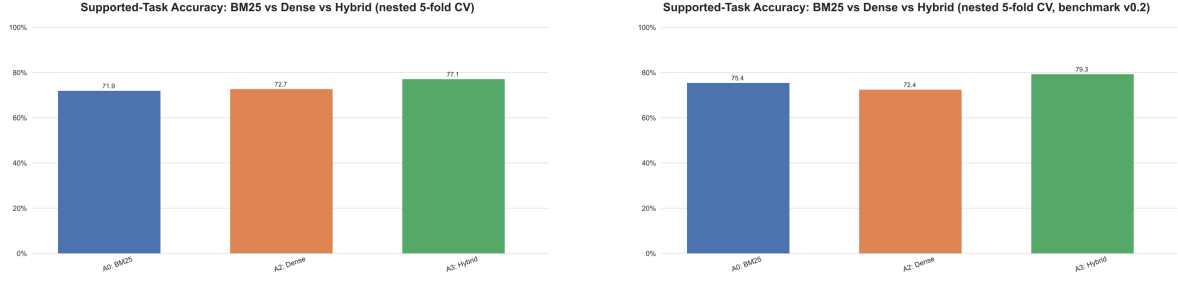


Figure 1: Supported-task accuracy for BM25, dense-only, and hybrid retrieval under nested 5-fold CV. Left: v0.1. Right: v0.2. The BM25→hybrid gain is similar in size on both versions; its exact-test significance differs (Table 1) and rests on a few queries (Table 2).

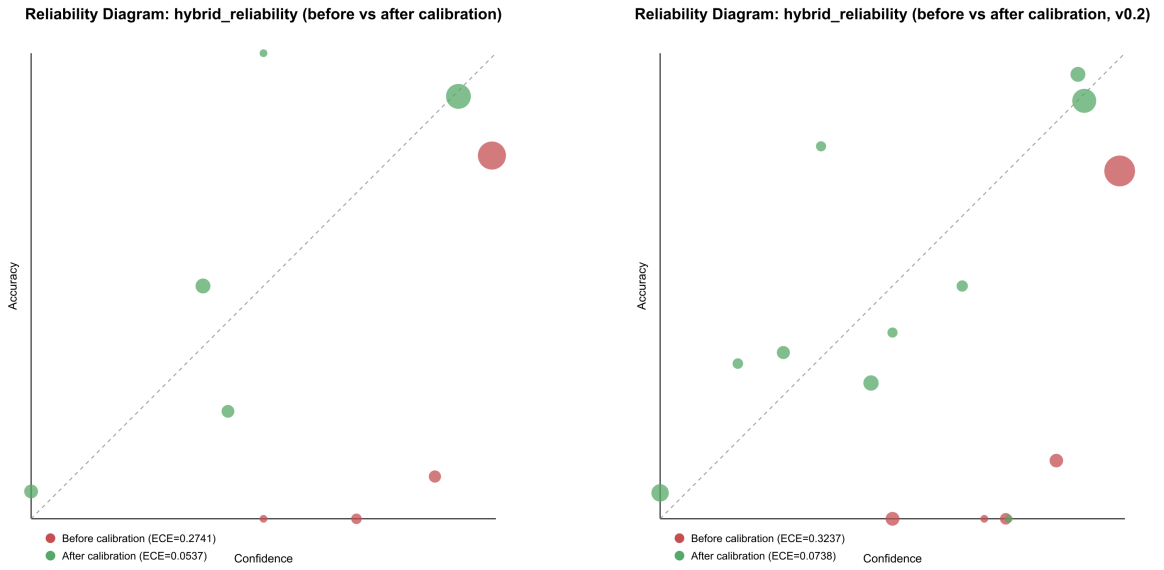


Figure 2: Reliability diagram for the hybrid system's fused-score confidence, before vs. after isotonic calibration. Left: v0.1. Right: v0.2.

**Intent-held-out generalization (Split B).** GroupKFold over 171 derived intent groups (v0.2; 0 groups split across folds, verified programmatically) shows that hybrid accuracy and OOD detection do not depend on intent overlap between tuning and test: non-OOD accuracy is 79.24% under Split B vs. 79.25% under Split A, and OOD AUROC is 0.898 vs. 0.901 — both effectively unchanged. Calibration, while still a large improvement, is measurably **attenuated on held-out intents**: a 69.0% vs. 77.2% relative ECE reduction. The identical protocol on v0.1 (125 groups) shows the same qualitative pattern at smaller, noisier scale (Table 3), consistent with the small v0.1 OOD subset.

## 7 Ablation, Functional, and Safety Evaluation

**Ablation (v0.1 → v0.2).** A0 (BM25) = 71.9%/75.4%; A1 (BM25, no substring bonus) is **identical** to A0 on both benchmarks (0 command differences on 150 or 209 queries) — the bonus fires on 47/150 v0.1 queries but never changes the top-1 winner, a null result we report plainly rather than let the bonus's presence imply an untested effect. A2 (dense) = 72.6–72.7%/72.4%; A3 (hybrid, nested-CV) = 77.0–77.1%/79.3%. A4 (+margin rejection) reaches 88.5% selective accuracy @ 70.8% coverage on v0.1; A5 (+OOD detection) reaches 90.1% @ 69.5% coverage (10/15 OOD caught). On v0.2, A5 is *lower* at *lower* coverage (82.2% @ 58.9%, 38/50 OOD caught) — the harder v0.2 query mix making the accept/reject decision more difficult, not a

| BM        | Evaluated on            | n   | BM25→Hyb.    | Dense→Hyb.   | ECE red. |
|-----------|-------------------------|-----|--------------|--------------|----------|
| v0.1      | all queries             | 135 | +5.2 (0.016) | +4.4 (0.146) | 80%      |
| v0.1      | w/o 9 bare-keyword      | 126 | +5.6 (0.016) | +4.8 (0.146) | 80%      |
| v0.2      | all queries             | 159 | +3.8 (0.070) | +6.9 (0.013) | 77%      |
| v0.2      | w/o 17 new bare-keyword | 142 | +4.9 (0.016) | +5.6 (0.057) | 81%      |
| v0.2      | w/o all 26 bare-keyword | 133 | +5.3 (0.016) | +6.0 (0.057) | 84%      |
| v0.1/v0.2 | answerable only         | 121 | +5.8 (0.016) | +4.1 (0.227) | 78/81%   |

Table 2: Sensitivity to bare-keyword ambiguous queries (post hoc). Cells show the hybrid’s accuracy gain in points over BM25 and over dense retrieval, with the exact McNemar  $p$  (unadjusted) in parentheses; ECE red. is the relative ECE reduction from isotonic calibration re-fit on the subset;  $n$  counts non-OOD queries evaluated. Fusion weight and thresholds were not re-tuned.

| BM   | Metric       | Split A   | Split B   | $\Delta$ |
|------|--------------|-----------|-----------|----------|
| v0.1 | Non-OOD acc. | 77.06%    | 77.04%    | −0.02pp  |
| v0.1 | OOD AUROC    | 0.867     | 0.849     | −0.018   |
| v0.1 | Ambig. F1    | 0.310     | 0.300     | −0.010   |
| v0.1 | Calib. ECE   | .274→.054 | .274→.065 | +0.011   |
| v0.2 | Non-OOD acc. | 79.25%    | 79.24%    | −0.01pp  |
| v0.2 | OOD AUROC    | 0.901     | 0.898     | −0.003   |
| v0.2 | Ambig. F1    | 0.496     | 0.479     | −0.017   |
| v0.2 | Calib. ECE   | .324→.074 | .324→.101 | +0.027   |

Table 3: Split A vs. Split B (intent-held-out generalization).

regression in the method. A6 (+calibration) makes the same decisions as A5 but with ECE 0.054 vs. 0.274 raw (Figure 5, Appendix A).

**Functional evaluation.** Sandboxed execution (disposable temp directories, never the host machine) on a deliberately narrow, explicitly-scoped 15/150-query subset (git + filesystem, placeholder-free, non-interactive, host-safe gold commands only). Gold commands: 100% functional success (validates benchmark quality). Retrieved (hybrid) commands: 93.3% functional success, zero cases of textually-correct-but-functionally-broken — a positive result that covers only 10% of the benchmark by design.

**Safety evaluation.** A fully deterministic, rule-based classifier (LOW/MEDIUM/HIGH/CRITICAL), authored from general domain knowledge before inspecting per-query outcomes, evaluated against the benchmark’s pre-existing risk-level ground truth: 89.6% exact 4-tier accuracy (112/125 gold commands), 95%/95% precision/recall on the actionable risky-vs-not-risky binary, zero dangerous-direction misses (no CRITICAL/HIGH command ever tagged safe). On retrieved commands, accuracy holds (90.0%) but risky-binary precision drops to 0.826 — a retrieval-error artifact, not a classifier

defect. On v0.2, exact accuracy and recall shift modestly downward (83.9%/0.864) because the classifier — deliberately not re-tuned against new examples — was authored before v0.2’s new gold commands existed, surfacing the same documented gaps, not new ones.

## 8 Blind Re-Annotation of a Sample of AI-Authored Queries

v0.2’s 59 new queries were authored and adjudicated by an AI agent under human direction. To check for systematic labeling error, we drew a stratified, seed-42 sample of 14 of them (8 OOD-labeled, 6 ambiguous-labeled) and had one reviewer judge each query fresh, blind to its original label, as clearly answerable, ambiguous, or OOD. The reviewer is the project’s human director, who directed the original authoring at a summary level but had not seen these 14 labels; the check is therefore blind but only partially independent, and is not the two-annotator protocol we intend to run (§10). Cohen’s  $\kappa = 0.6316$ , below our pre-specified 0.7 target. Agreement was perfect on the 8 OOD-labeled items (8/8), and all three disagreements were among the 6 ambiguous-labeled ones: “ffm-peg” and “pip” (single-token tool names) and “list running processes” (a short category phrase), each of which we had labeled ambiguous but the reviewer judged to have one obvious default command. The other three ambiguous items (“copy a file”, “security”, “system”) were confirmed. With  $n = 14$  the estimate is coarse: one flipped item would move it materially. The OOD labels, on which the OOD-rejection result rests, were fully confirmed; the ambiguous labels are the uncertain ones, and this is why §6 reports how the conclusions change when ambiguous queries are excluded. Note that the disputed items are not all single-token, so the “answerable only” rows of Table 2, which drop every ambiguous query, are the ones free of

this label uncertainty.

## 9 Discussion

The clearest result concerns reliability, not accuracy. Post-hoc isotonic calibration removes most of the baseline’s overconfidence (56.7–84.1% relative ECE reduction). It is the only compared effect that is significant after correction on both benchmark versions, it survives intent-held-out evaluation with some attenuation (69–77% vs. 76–80%), and it is unchanged when bare-keyword or all ambiguous queries are removed (Table 2). OOD rejection is the second solid result, but only once the OOD subset grew from 15 to 50 queries; its labels are the ones our re-annotation fully confirmed. The accuracy gain over BM25 is real in size but not established in significance: it is similar on both versions (+5.2 and +3.8 points), and every discordant query but one favors the hybrid, yet the exact test has only 7–8 discordant queries to work with, so a single query (“system”) moves it across the threshold. The hybrid-over-dense result on v0.2 depends on three new bare-keyword queries. We draw two lessons. First, at this benchmark size, significance statements about accuracy differences are fragile to individual queries, the power concern raised by Card et al. (2020); effect sizes with intervals are more informative than  $p$ -values here. Second, v0.2’s additional queries are all OOD or ambiguous, so they test detection and calibration but add no answerable evidence: expanding the benchmark helped the OOD question and did not help the accuracy question. Ambiguity detection remains weak, and our re-annotation suggests part of the difficulty is that the ambiguous label itself is unsettled for short queries.

## Limitations

- **Benchmarks are not independent, and v0.2 adds no answerable evidence.** v0.2 contains all of v0.1; the 121 answerable queries are identical in both versions, and every difference comes from added OOD and ambiguous queries. Results on the two versions are not independent replications, and their Holm families are not independent either.
- **Fragile accuracy inference.** The exact tests on accuracy rest on 7–8 discordant queries; one query decides whether BM25→hybrid is significant on v0.2, and three decide hybrid-

vs.-dense (§6). We treat the accuracy gain as suggestive.

- **Exploratory statistics.** The Holm family and the choice of calibration as the primary hypothesis were fixed after the raw results were known, and the sensitivity analysis is post hoc (fusion weight and thresholds were not re-tuned on the reduced sets). No result here is confirmatory.
- **Label uncertainty in AI-authored queries.** v0.2’s 59 new queries were authored and adjudicated by an AI agent under human direction. A blind re-annotation of 14 by one partially independent reviewer gave  $\kappa = 0.63$  (§8): OOD labels were confirmed, ambiguous labels were not (3/6 disagreed). We have not run a two-annotator protocol.
- **Ambiguity detection is weak** (F1 0.31–0.50 under both class-stratified and intent-held-out evaluation), and the ambiguous label itself is unsettled for short queries.
- **Canonical-query independence.** All 25 canonical queries are verbatim-equal to a corpus intent, so 100% canonical accuracy is a positive-control sanity check, not evidence of language understanding; the paraphrase, low-overlap, polysemy, and ambiguous types carry the capability claims.
- **Ground-truth defects.** Three items have a gold or acceptable command that is not a record of the win32 corpus the system searches: TA-B187 (“tar”; gold and all three acceptable commands exist only in the Linux/macOS records), TA-B145 (gold absent; already classed as needing correction), and TA-B149 (one acceptable command absent). None affects a comparison between systems (every system misses TA-B145 and TA-B187 and hits TA-B149), but the first two lower every system’s absolute accuracy by about 1 point on v0.2 (0.6 on v0.1). We report both versions unchanged; the items should be corrected or dropped in a later version.
- **Corpus scale and platform.** 279 win32-visible unique intents (431 raw records), small relative to generation-oriented corpora and single-platform; the planned 500–800-intent expansion is not done.

- **Calibration under intent shift.** The ECE reduction attenuates from 77% within-distribution to 69% on intent-held-out evaluation, and isotonic calibration is sensitive to small samples.
- **Evaluation scope.** No human-preference study; a fixed embedding model; functional evaluation covers only 15 of 150 queries and treats success as exit code 0, not verified side effects; the safety classifier is evaluated against the benchmark’s own risk labels.
- **Literature coverage.** Our literature search was limited, so we make no claim of priority or novelty beyond the comparisons cited.

## Ethics Statement

TermAssist retrieves from a fixed, human-curated command library rather than generating arbitrary shell commands, which structurally bounds the space of possible harmful outputs to commands a human already vetted — but a retrieved command can still be inappropriate for a given user’s intent, and our own baseline audit found the deployed system is frequently *confidently* wrong. We evaluate command risk with a separate, deterministic safety classifier (§7) specifically because retrieval correctness and command risk are orthogonal concerns; a correctly-retrieved command can still be destructive if misapplied, and no component in this study executes a retrieved command against a user’s real system. All functional-evaluation command execution in this study ran in disposable, sandboxed temporary directories, never against a host machine. We disclose the AI-assisted authorship of part of our benchmark (§3) and the below-target result of our own partially independent re-annotation of it (§8) rather than omit or minimize either.

## 10 Conclusion and Future Work

A lightweight, fully offline, non-LLM hybrid retriever with post-hoc isotonic calibration substantially improves the reliability of a shipped BM25 command-retrieval baseline. The most defensible results are the calibration improvement, which holds on both benchmark versions, on held-out intents, and when ambiguous queries are removed, and the OOD-rejection improvement, which is significant on the larger OOD subset. The hybrid’s accuracy gain over BM25 is consistent in size but statistically fragile, the hybrid-over-dense advantage

on v0.2 depends on a few bare-keyword queries, and ambiguity detection is weak. All significance analyses are exploratory. Future work, in priority order: (1) a written annotation codebook for when a short query counts as ambiguous, followed by two independent annotators labeling all 59 AI-authored queries, with  $\kappa$ , adjudication, and a frozen benchmark; (2) a confirmatory analysis specified before running on that benchmark; (3) more answerable queries, since v0.2 adds none; (4) an independent human-labeled safety set not derived from the classifier’s own logic; (5) expanding the corpus from 279 to roughly 500–650 human-validated intents, and functional evaluation beyond 15 tasks; and (6) a human-preference study. Items (1)–(5) need human effort and cannot responsibly be replaced by bulk automated generation.

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## **A Additional Figures**

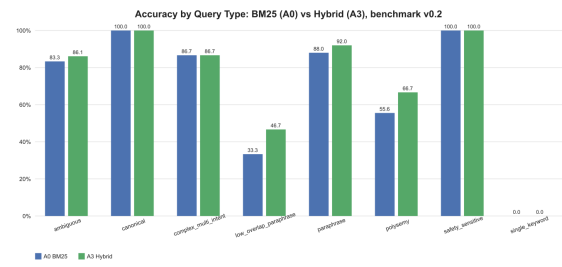
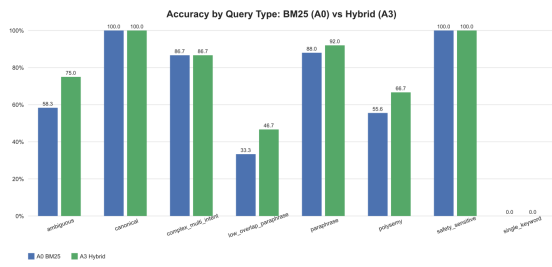


Figure 3: Accuracy by query type, BM25 (A0) vs. hybrid (A3). Left: v0.1. Right: v0.2.

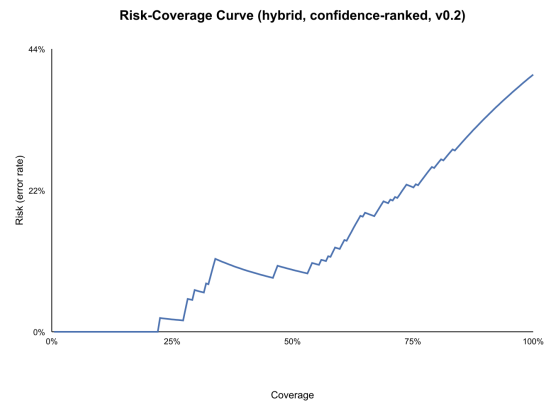
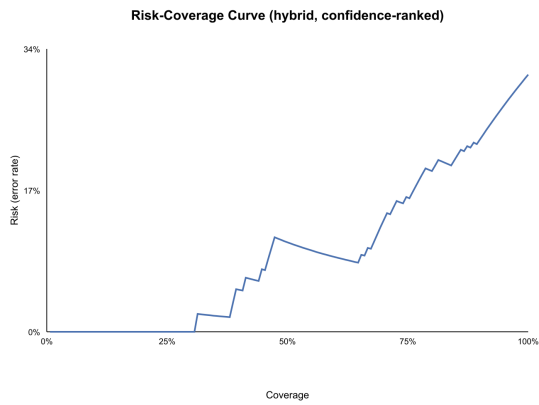


Figure 4: Risk-coverage curve for the hybrid system under confidence-ranked selective answering. Left: v0.1. Right: v0.2.

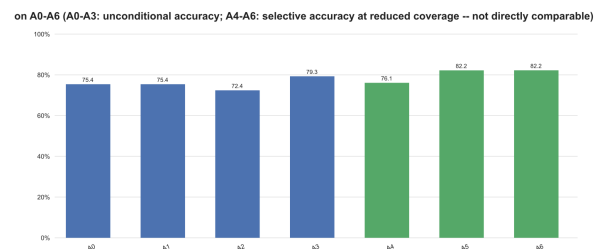
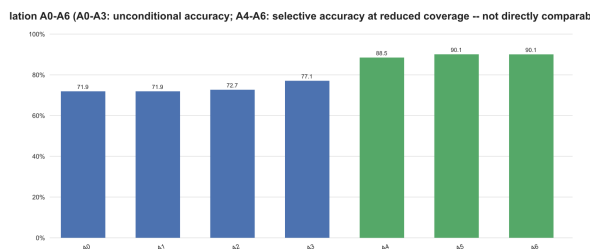


Figure 5: Ablation A0–A6. A0–A3 report unconditional accuracy; A4–A6 report selective accuracy at reduced coverage and are not directly comparable to A0–A3’s bars. Left: v0.1. Right: v0.2.